SOLAR PV DESIGN PORTFOLIO

Four Solar PV Systems for Bangladesh

Industrial rooftop | Hybrid with battery | Solar carport with EV charging | Solar irrigation pump

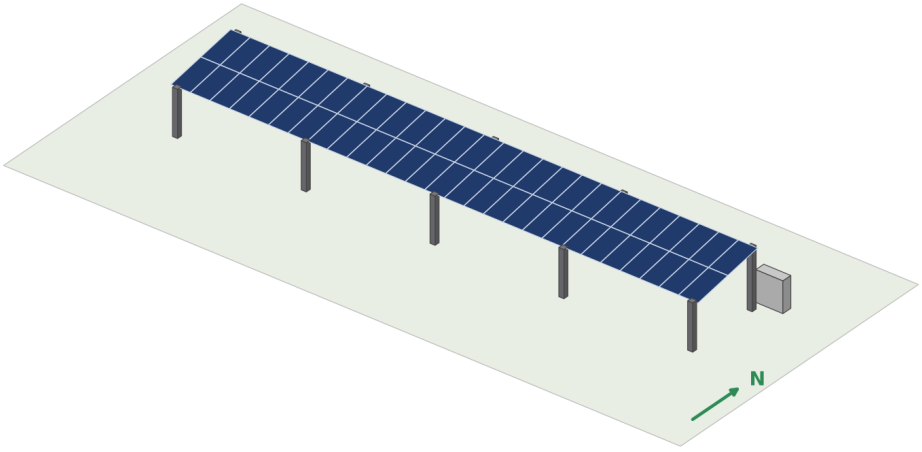
Layout and single-line diagrams (AutoCAD), design calculations (Excel), 3D models (SketchUp) and hourly energy simulation (Python / pvlib) validated against PVsyst

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Portfolio summary

This portfolio presents four grid-connected and off-grid photovoltaic designs for typical Bangladeshi applications. Each design covers the array layout, string sizing against inverter limits, a single-line diagram with protection and earthing, a bill of quantities, a 3D site model, and an hourly energy simulation for a full year, followed by an analysis specific to the application.

#	Project	Capacity	Energy (kWh/yr)	kWh/kWp	PR	Key result
2	49.5 kWp industrial rooftop, Gazipur	49.50 kWp	60,756	1,227	0.77	95% of PV used on site
3	4.95 kWp hybrid + battery, Uttara	4.95 kWp	6,588	1,331	0.80	93% self-sufficiency
4	29.7 kWp solar carport + EV charging	29.70 kWp	39,278	1,322	0.81	92% of EV energy from solar (scheduled)
5	8.25 kWp solar irrigation pump, Rajshahi	8.25 kWp	11,263	1,365	0.80	55,853 m3 water per year



29.7 kWp solar carport with EV charging, AUST campus (Project 4).

Design method and simulation model

Design workflow

1. **Site and load data:** roof or plot geometry, obstructions, load profile and Meteonorm 9.0 climate data for the site region.

2. **Array layout:** module placement with setbacks; row pitch from the 21 December 09:00 sun position so rows do not shade each other between 09:00 and 15:00.

3. **Electrical design:** string length from the cold-weather open-circuit voltage and hot-weather MPP voltage; MPPT current and DC/AC ratio checks; cable sizing for under 1.5% DC and 1% AC voltage drop; protection and earthing.

4. **Drawings and model:** layout plan and single-line diagram in AutoCAD (DXF); 3D site model with shadow-study scenes in SketchUp.

5. **Energy simulation:** hourly simulation for one year, loss breakdown, and comparison with a hand calculation.

6. **Application analysis:** on-site use, battery dispatch, EV charging coverage or pumped water, depending on the system.

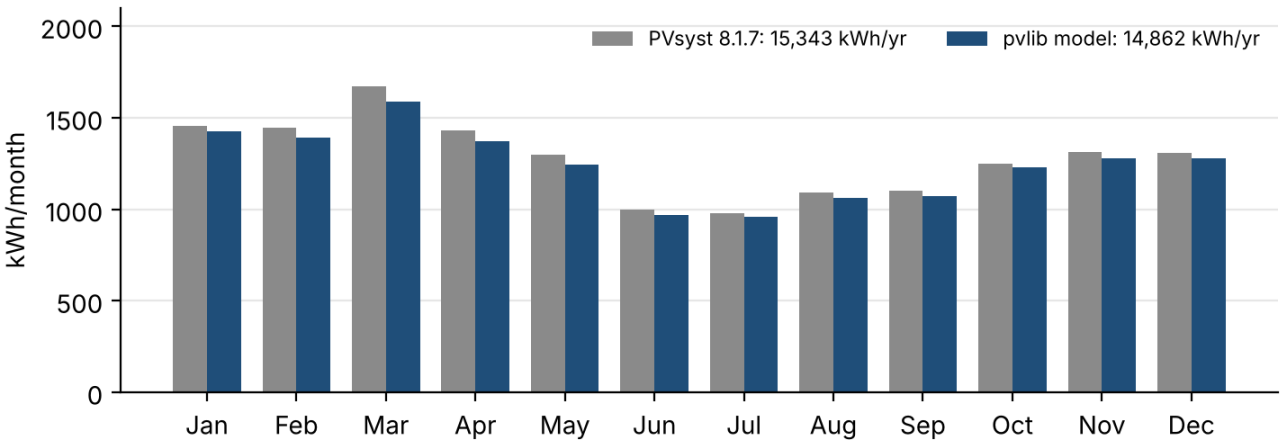
Simulation model

The hourly model is written in Python with pvlib 0.16: Perez transposition to the plane of array, a physical incidence-angle model for glass reflection, SAPM cell temperature (open-rack or close-roof mounting), the PVWatts DC model with a temperature coefficient of -0.35%/C, a loss chain for soiling, shading, mismatch, wiring, connections and light-induced degradation, and the PVWatts inverter model including clipping. Weather is a synthetic hourly year built from Meteonorm 9.0 monthly global and diffuse irradiation and temperature for Dhaka, using clear-sky profiles, seeded day-to-day variability and exact monthly totals; a PVGIS typical year is used instead when available.

Validation against PVsyst

The model was checked against a PVsyst 8.1.7 simulation of a reference 11 kWp rooftop system in Mirpur (20 x 550 Wp, 23 deg tilt, 9 kW inverter, 3D near-shading scene).

Quantity	PVsyst 8.1.7	pvlib model	Difference
Irradiation on plane (kWh/m2/yr)	1,679	1,696	+1.0%
Energy to grid (kWh/yr)	15,343	14,862	-3.1%
Specific yield (kWh/kWp/yr)	1,395	1,351	-3.1%
Performance ratio	0.831	0.797	-0.034



Monthly energy of the reference system: PVsyst and the pvlib model agree within about 3% annually. The model is slightly conservative because its default loss chain is more pessimistic than PVsyst's.

Project 2: 49.5 kWp Industrial Rooftop PV - Garment Factory, Gazipur

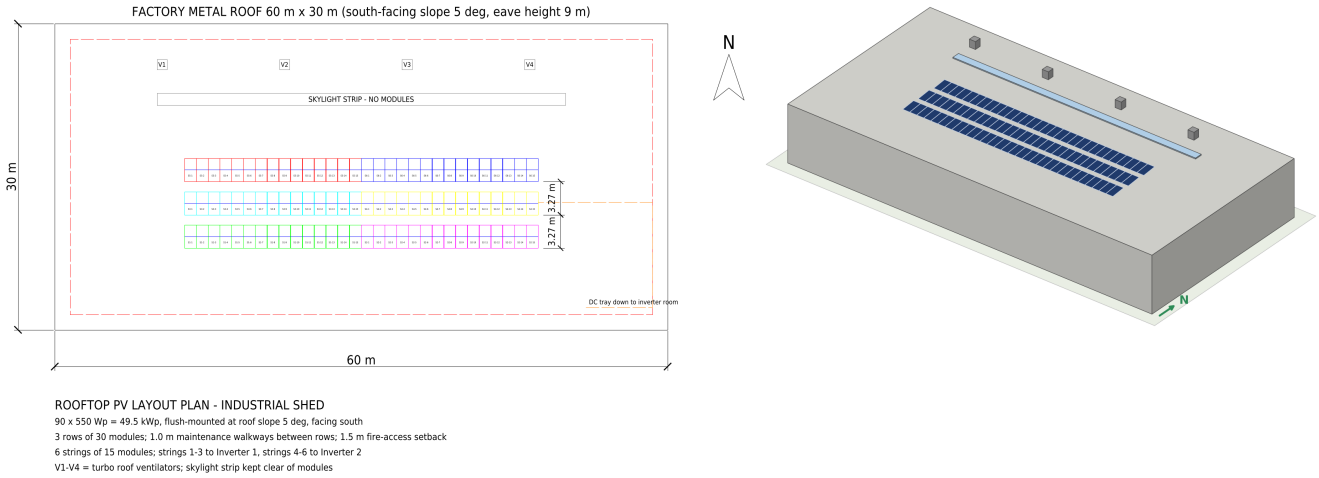
A first-phase rooftop system for a garment factory with a large south-facing metal shed roof. The modules are flush-mounted on rails at the 5 deg roof slope, which avoids tilt frames, reduces wind load and keeps roof penetrations to a minimum. Two string inverters feed the factory's LT panel through an AC combiner panel.

49.50 kWp	60,756	1,227	0.77	95%
90 x 550 Wp	kWh per year (simulated)	kWh/kWp per year	performance ratio	used on site

1. Site and design basis

Parameter	Value
Location	Gazipur, Bangladesh (23.99 N, 90.42 E)
Site	Factory metal roof 60 m x 30 m (south-facing slope 5 deg, eave height 9 m)
Orientation	Tilt 5 deg, azimuth 0 deg (south); roof slope 5 deg; modules flush-mounted on rails (no tilt frames)
PV array	90 x 550 Wp mono half-cell = 49.50 kWp; 6 string(s) of 15
Inverter / controller	2 x 20 kW three-phase string inverters, 2 MPPT each
Design temperatures	Minimum ambient 8 C; maximum cell 75 C
Weather data	Meteonorm 9.0 monthly values (Dhaka region), hourly synthetic year

2. Array layout and 3D model

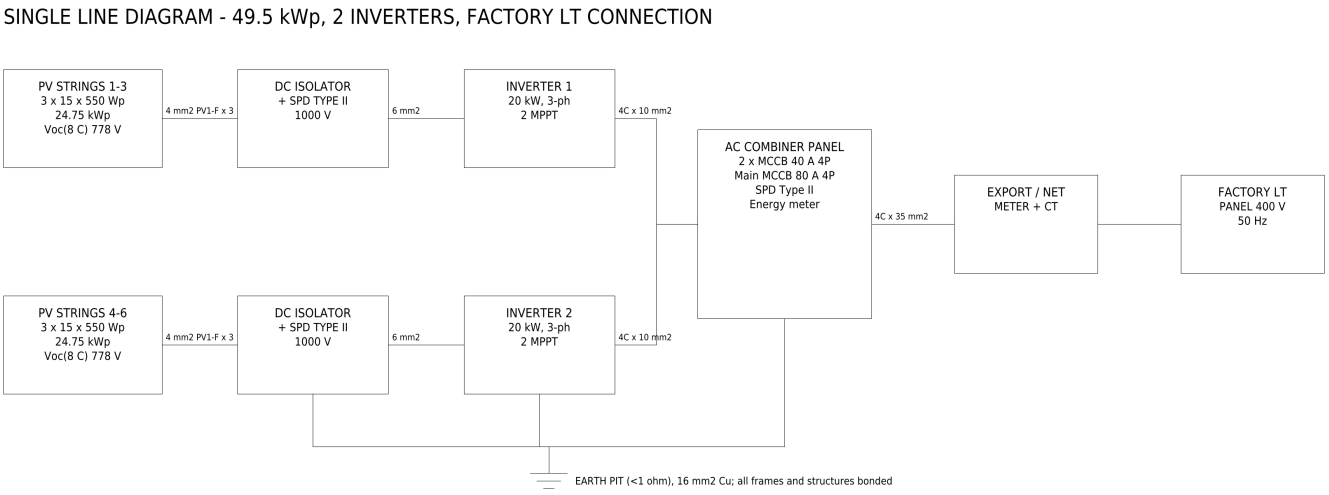


Layout plan from drawing PV-02 (left; string groups in separate colours) and 3D site model (right).

90 x 550 Wp = 49.5 kWp, flush-mounted at roof slope 5 deg, facing south. 3 rows of 30 modules; 1.0 m maintenance walkways between rows; 1.5 m fire-access setback. 6 strings of 15 modules; strings 1-3 to Inverter 1, strings 4-6 to Inverter 2. V1-V4 = turbo roof ventilators; skylight strip kept clear of modules.

3. Electrical design

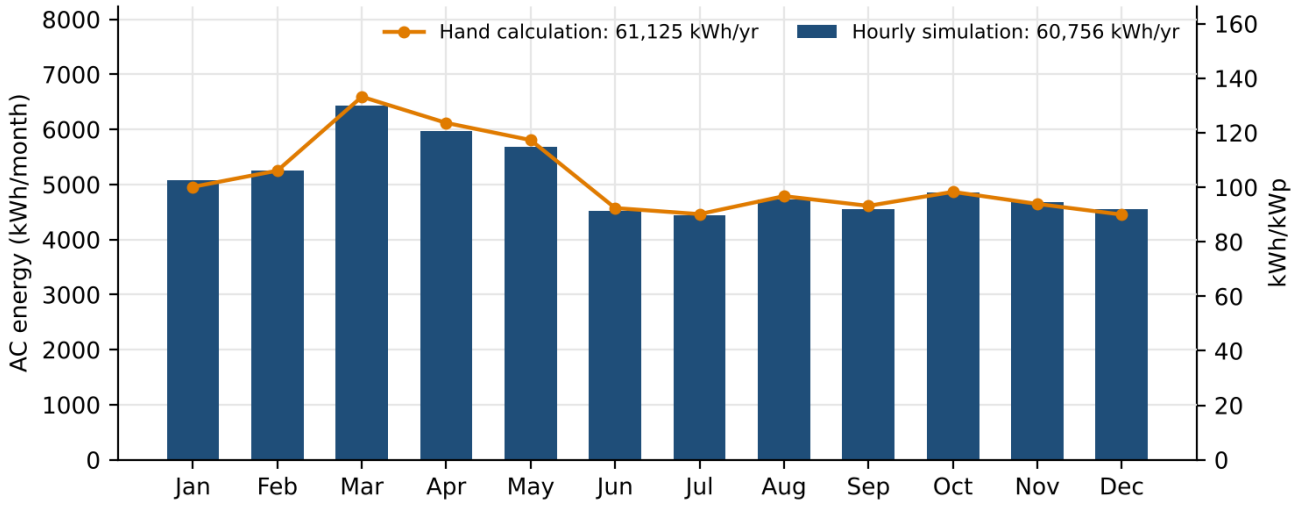
Check	Calculation	Result	Limit	Status
Max string voltage (8 C)	15 x 49.6 V x [1 + (-0.27%/C)(8 - 25)]	778 V	max 1100 V	OK
Min MPP voltage (cell 75 C)	15 x 41.9 V x [1 + (-0.35%/C)(75 - 25)]	519 V	min 200 V	OK
MPP voltage at STC	15 x 41.9 V	628 V	max 1000 V	OK
MPPT input current	2 string(s) x 13.13 A	26.3 A	max 32 A	OK
DC/AC ratio	49.50 kWp / 40.0 kW	1.24	1.0 - 1.3	OK



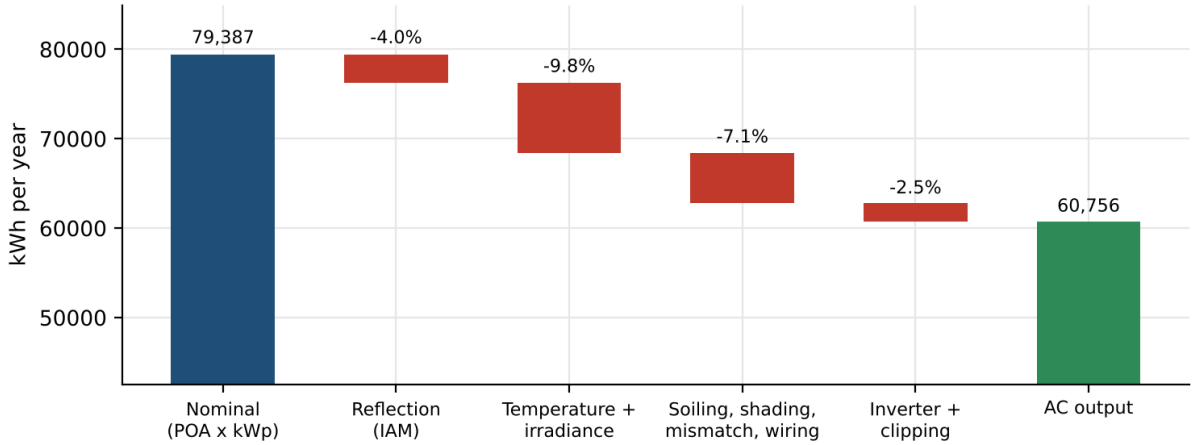
Single-line diagram, drawing PV-02.

Inverters and AC combiner panel in ground-floor electrical room. Roof cable tray with UV-rated DC cable. Export limitation / zero-export relay if required by the utility. Ratings to be verified with selected datasheets.

4. Energy simulation

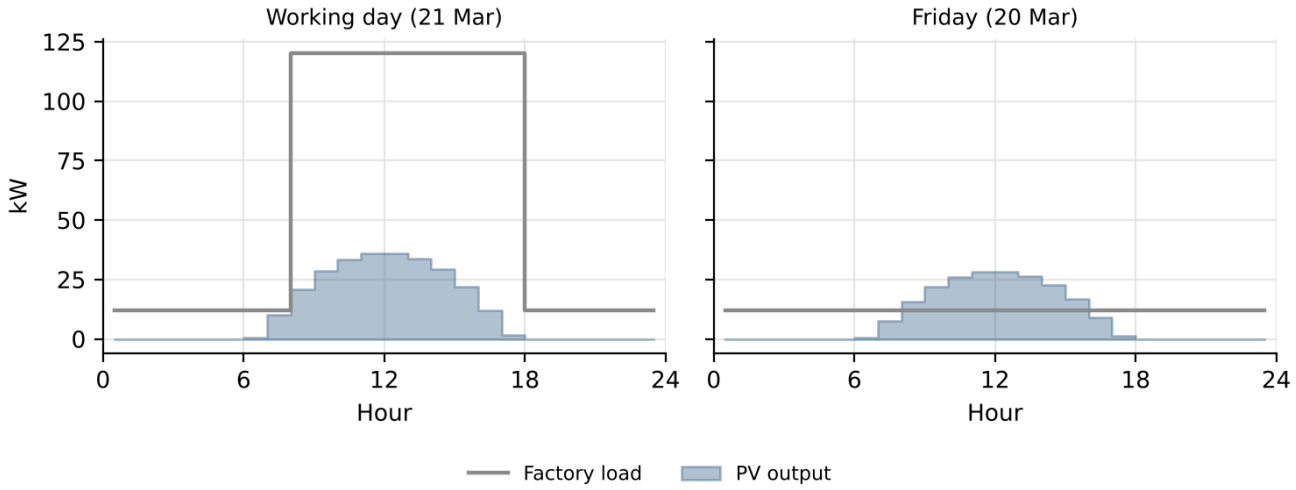


Monthly AC energy: hourly simulation 60,756 kWh/yr against a hand calculation of 61,125 kWh/yr (-0.6%). The hand calculation uses a fixed PR of 0.78; the simulation resolves each loss hour by hour.



Annual loss breakdown. Temperature and low-irradiance losses are 10.2% of effective irradiation; inverter clipping is 227 kWh/yr; peak AC output 38.0 kW.

5. On-site consumption



PV output against a 120 kW daytime factory load (08:00-18:00, Saturday to Thursday).

The factory absorbs 95% of the PV energy; only Friday production (3,175 kWh/yr) is exported. The array covers 13% of the annual factory consumption, so the roof, which has space for several times this capacity, can host a larger second phase with little export. Flush mounting at 5 deg yields 60,756 kWh/yr against 67,102 kWh/yr for an open-rack array at 23 deg (-9.5%); this yield is traded for no tilt frames, lower wind uplift and fewer roof penetrations on the metal sheet.

6. Bill of quantities

#	Item	Specification	Qty
1	PV module	550 Wp mono half-cell, IEC 61215/61730	90 nos
2	String inverter	20 kW, 3-phase, 2 MPPT, IEC 62109/61727	2 nos
3	Roof mounting rails + clamps	Aluminium rail on trapezoidal metal roof, flush 5 deg	49.5 kWp
4	DC isolator + SPD box	1000 V DC isolator, Type II SPD, per inverter	2 set
5	DC solar cable	4 mm2 PV1-F, red + black (6 strings x 2 x 60 m)	720 m
6	MC4 connectors	1000 V, IP67	16 pair
7	AC cable	4C x 10 mm2 Cu XLPE, inverters to AC combiner	30 m
8	AC cable	4C x 35 mm2 Cu XLPE, AC combiner to LT panel	40 m
9	AC combiner panel	2 x MCCB 40 A, main MCCB 80 A, SPD, energy meter	1 set
10	Export / net meter + CTs	As per utility (REB/DESCO) requirement	1 set
11	Cable tray	GI perforated tray with cover, roof + riser	80 m
12	Roof walkway	FRP grating walkway for O&M;	100 m
13	Earthing	Earth pits + 16 mm2 Cu, rails bonded	3 set
14	Lightning protection	Air terminals + down conductors	2 set
15	Installation, testing & commissioning	Labour, testing per IEC 62446	1 lot

Project 3: 4.95 kWp Hybrid Rooftop PV with Battery Backup - Uttara, Dhaka

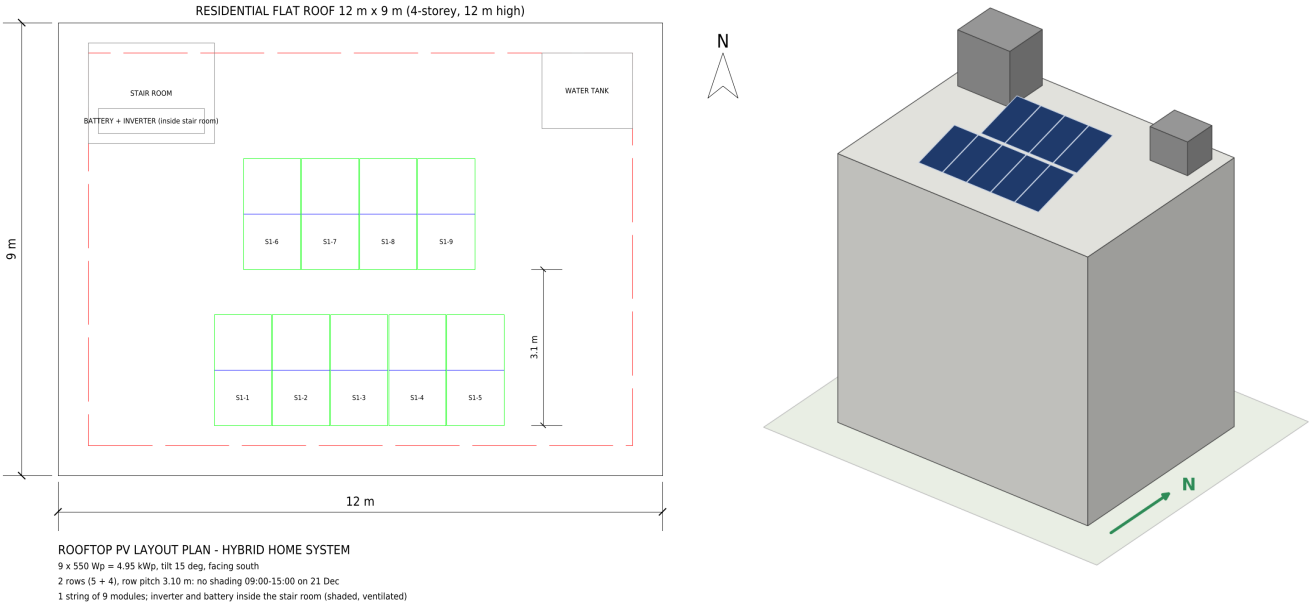
A residential system for a four-storey house in Uttara, designed for daily load-shedding. A hybrid inverter supplies the home, charges a LiFePO4 battery and exports surplus energy through a net meter; during grid outages it keeps a 1.2 kW critical-load board running from PV and battery.

4.95 kWp	6,588	1,331	0.80	93%
9 x 550 Wp	kWh per year (simulated)	kWh/kWp per year	performance ratio	self-sufficiency

1. Site and design basis

Parameter	Value
Location	Dhaka, Bangladesh (23.87 N, 90.4 E)
Site	Residential flat roof 12 m x 9 m (4-storey, 12 m high)
Orientation	Tilt 15 deg, azimuth 0 deg (south); 15 deg: compact rows on a small roof, still good yield
PV array	9 x 550 Wp mono half-cell = 4.95 kWp; 1 string(s) of 9
Inverter / controller	5 kW single-phase hybrid inverter, 48 V battery
Design temperatures	Minimum ambient 8 C; maximum cell 70 C
Weather data	Meteonorm 9.0 monthly values (Dhaka region), hourly synthetic year

2. Array layout and 3D model

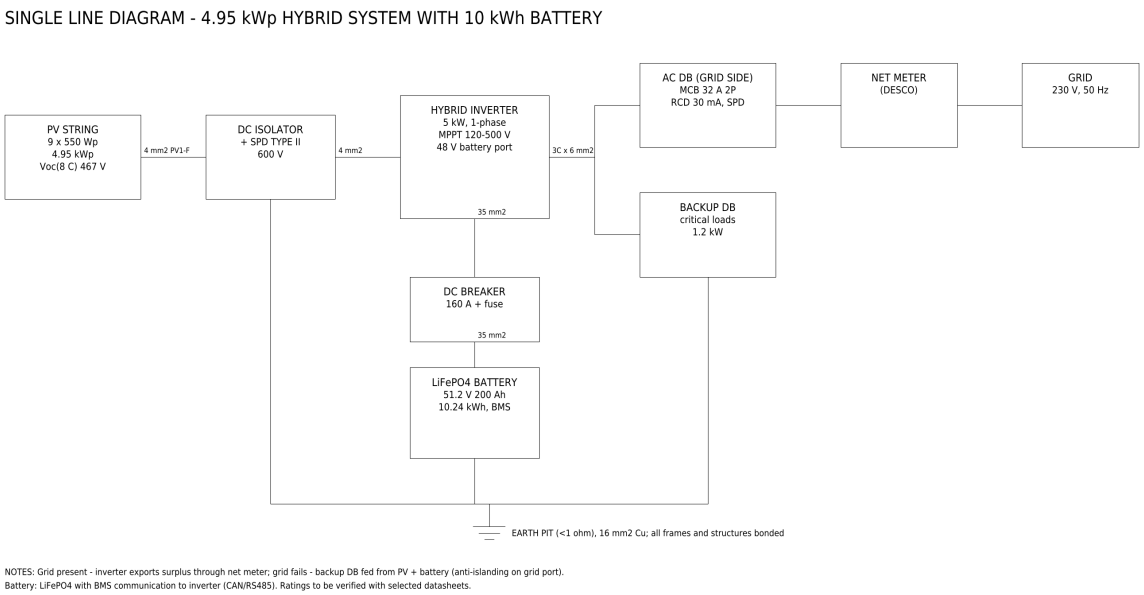


Layout plan from drawing PV-03 (left; string groups in separate colours) and 3D site model (right).

9 x 550 Wp = 4.95 kWp, tilt 15 deg, facing south. 2 rows (5 + 4), row pitch 3.10 m: no shading 09:00-15:00 on 21 Dec. 1 string of 9 modules; inverter and battery inside the stair room (shaded, ventilated).

3. Electrical design

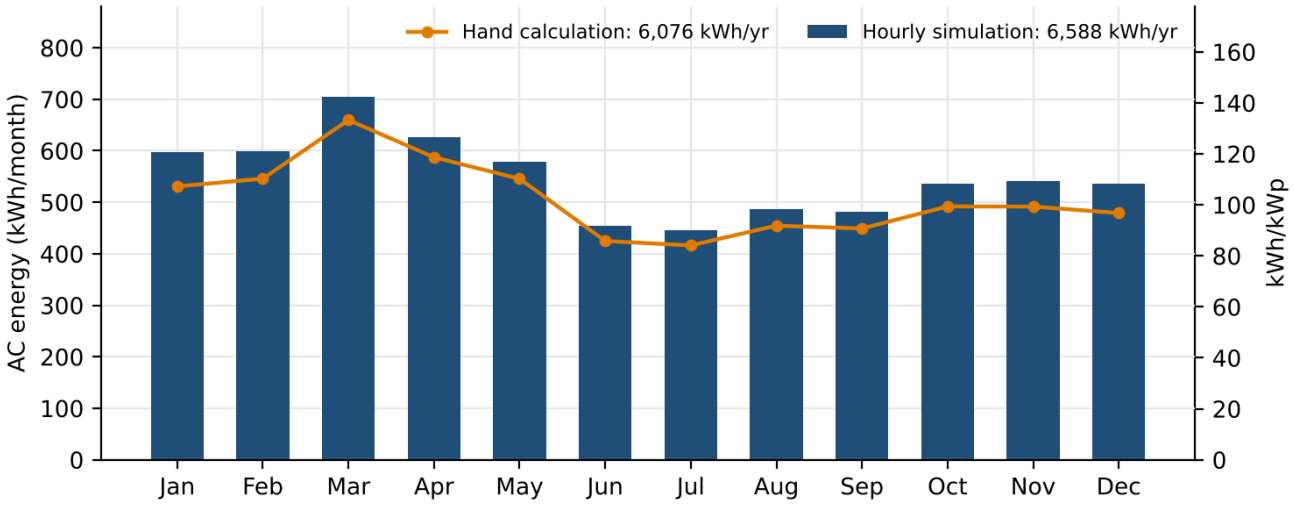
Check	Calculation	Result	Limit	Status
Max string voltage (8 C)	9 x 49.6 V x [1 + (-0.27%/C)(8 - 25)]	467 V	max 550 V	OK
Min MPP voltage (cell 70 C)	9 x 41.9 V x [1 + (-0.35%/C)(70 - 25)]	318 V	min 120 V	OK
MPP voltage at STC	9 x 41.9 V	377 V	max 500 V	OK
MPPT input current	1 string(s) x 13.13 A	13.1 A	max 16 A	OK
DC/AC ratio	4.95 kWp / 5.0 kW	0.99	0.9 - 1.3	OK



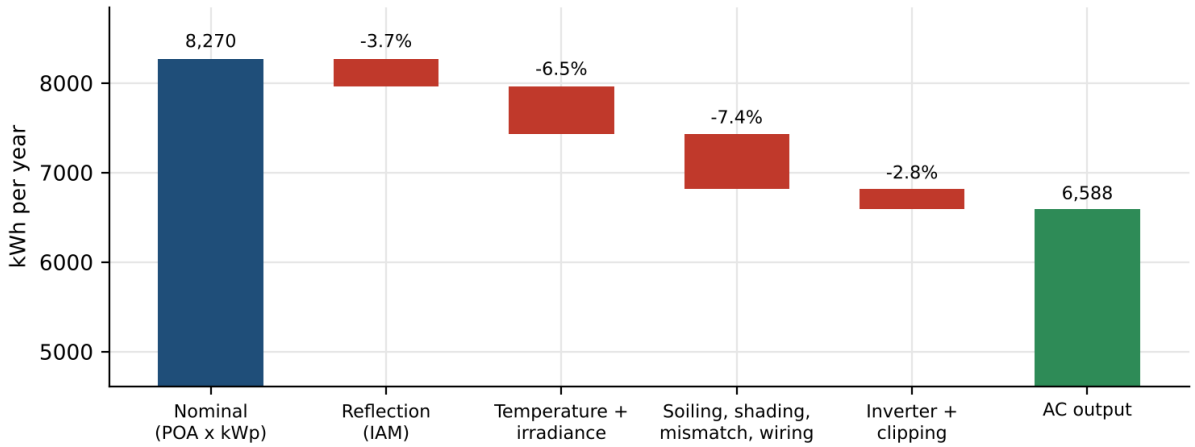
Single-line diagram, drawing PV-03.

Grid present - inverter exports surplus through net meter; grid fails - backup DB fed from PV + battery (anti-islanding on grid port). Battery: LiFePO4 with BMS communication to inverter (CAN/RS485). Ratings to be verified with selected datasheets.

4. Energy simulation

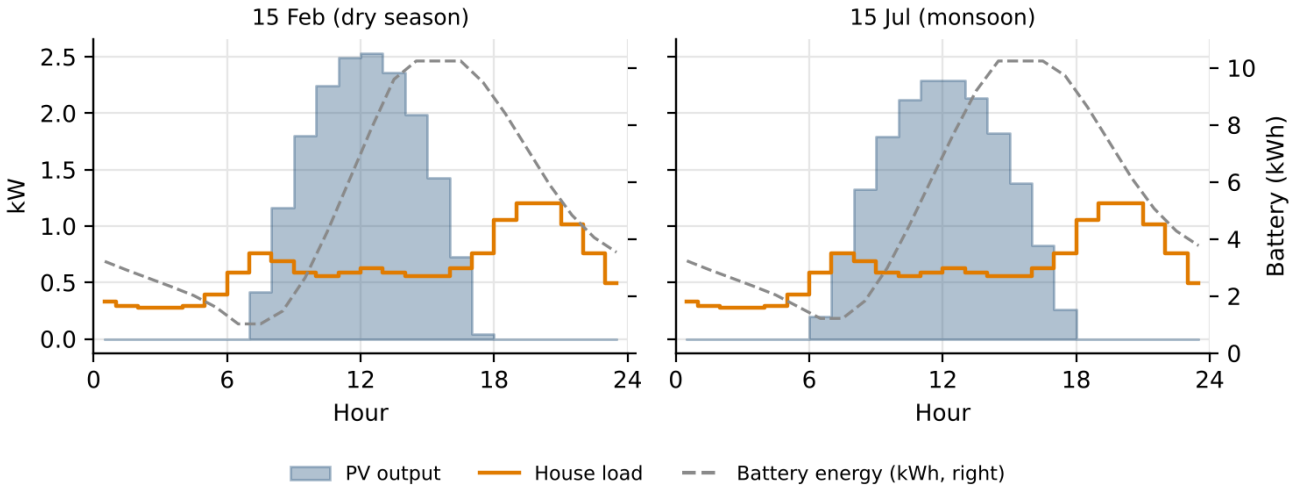


Monthly AC energy: hourly simulation 6,588 kWh/yr against a hand calculation of 6,076 kWh/yr (+8.4%). The hand calculation uses a fixed PR of 0.75; the simulation resolves each loss hour by hour.



Annual loss breakdown. Temperature and low-irradiance losses are 6.7% of effective irradiation; inverter clipping is 33 kWh/yr; peak AC output 4.2 kW.

5. Battery operation and backup



Hourly operation on a dry-season and a monsoon day: PV output, household load and stored battery energy.

Indicator (household 15 kWh/day, 10.24 kWh battery)	Value
PV energy used in the home (self-consumption)	81%
Household demand met by PV and battery (self-sufficiency)	93%
Energy exported / imported	1,225 / 334 kWh/yr
Equivalent full battery cycles	314 per year
Evening outage hours (19:00-21:00 daily) with critical load fully served	93% of 730 h

The battery is sized for 6 h of 1.2 kW critical load (7.2 kWh from the battery; 8.7 kWh nominal at 90% depth of discharge and 92% round-trip efficiency), so the selected 10.24 kWh unit covers the evening outage on most days; the uncovered hours fall after consecutive cloudy monsoon days.

6. Bill of quantities

#	Item	Specification	Qty
1	PV module	550 Wp mono half-cell, IEC 61215/61730	9 nos
2	Hybrid inverter	5 kW, 1-phase, 48 V, on/off-grid, IEC 62109	1 nos
3	Lithium battery	LiFePO4 51.2 V 200 Ah (10.24 kWh) with BMS	1 nos
4	Battery rack + DC breaker	160 A DC breaker, fuse, busbar	1 set
5	Mounting structure	HDG / aluminium, 15 deg fixed tilt	4.95 kWp
6	DC isolator + SPD	600 V DC isolator, Type II SPD	1 set
7	DC solar cable	4 mm2 PV1-F, red + black	40 m
8	Battery cable	35 mm2 flexible Cu with lugs	4 m
9	AC cable	3C x 6 mm2 and 3C x 4 mm2 Cu	30 m
10	AC DB (grid side)	MCB 32 A 2P, RCD 30 mA, Type II SPD	1 set
11	Backup DB	Critical-load DB with MCBs	1 set
12	Net meter	Bi-directional, as per DESCO	1 nos
13	Earthing	Earth pit + 16 mm2 Cu	2 set
14	Installation, testing & commissioning	Labour, testing per IEC 62446	1 lot

Project 4: 29.7 kWp Solar Carport with EV Charging - AUST Campus, Tejgaon

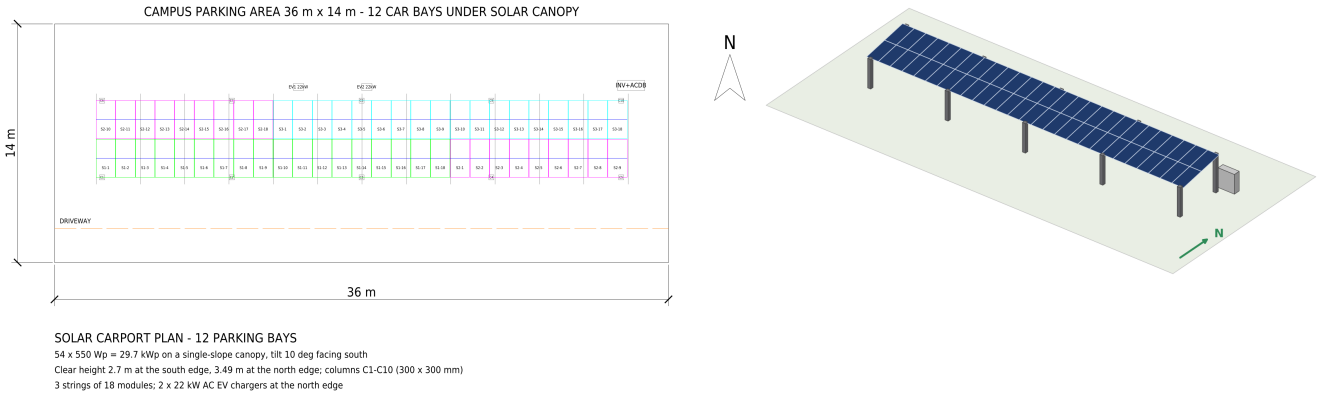
A single-slope solar canopy over 12 parking bays at the AUST campus, with two 22 kW AC chargers. The study quantifies how much EV charging energy the canopy can supply, and how scheduling charging sessions into solar hours changes that share, linking the design to research on PV-integrated EV charging scheduling.

29.70 kWp 54 x 550 Wp	**39,278** kWh per year (simulated)	**1,322** kWh/kWp per year	**0.81** performance ratio	**92%** EV energy from solar (scheduled)

1. Site and design basis

Parameter	Value
Location	Dhaka, Bangladesh (23.76 N, 90.41 E)
Site	Campus parking area 36 m x 14 m - 12 car bays under solar canopy
Orientation	Tilt 10 deg, azimuth 0 deg (south); 10 deg canopy: rain run-off, low wind load, enough clear height
PV array	54 x 550 Wp mono half-cell = 29.70 kWp; 3 string(s) of 18
Inverter / controller	25 kW three-phase string inverter, 2 MPPT
Design temperatures	Minimum ambient 8 C; maximum cell 65 C
Weather data	Meteonorm 9.0 monthly values (Dhaka region), hourly synthetic year

2. Array layout and 3D model

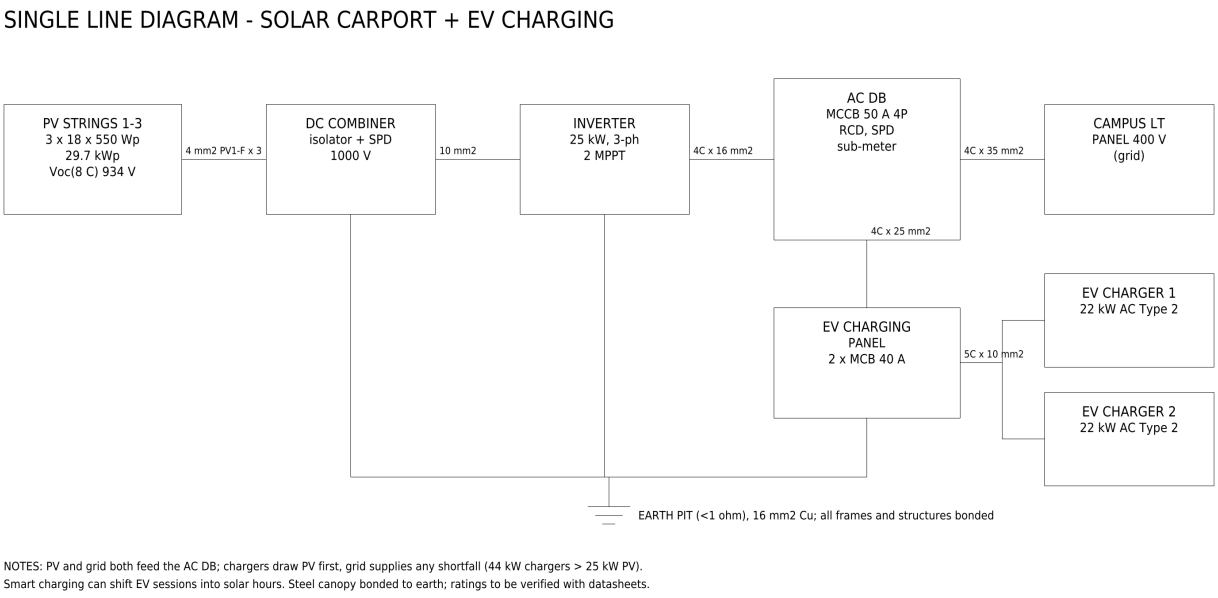


Layout plan from drawing PV-04 (left; string groups in separate colours) and 3D site model (right).

54 x 550 Wp = 29.7 kWp on a single-slope canopy, tilt 10 deg facing south. Clear height 2.7 m at the south edge, 3.49 m at the north edge; columns C1-C10 (300 x 300 mm). 3 strings of 18 modules; 2 x 22 kW AC EV chargers at the north edge.

3. Electrical design

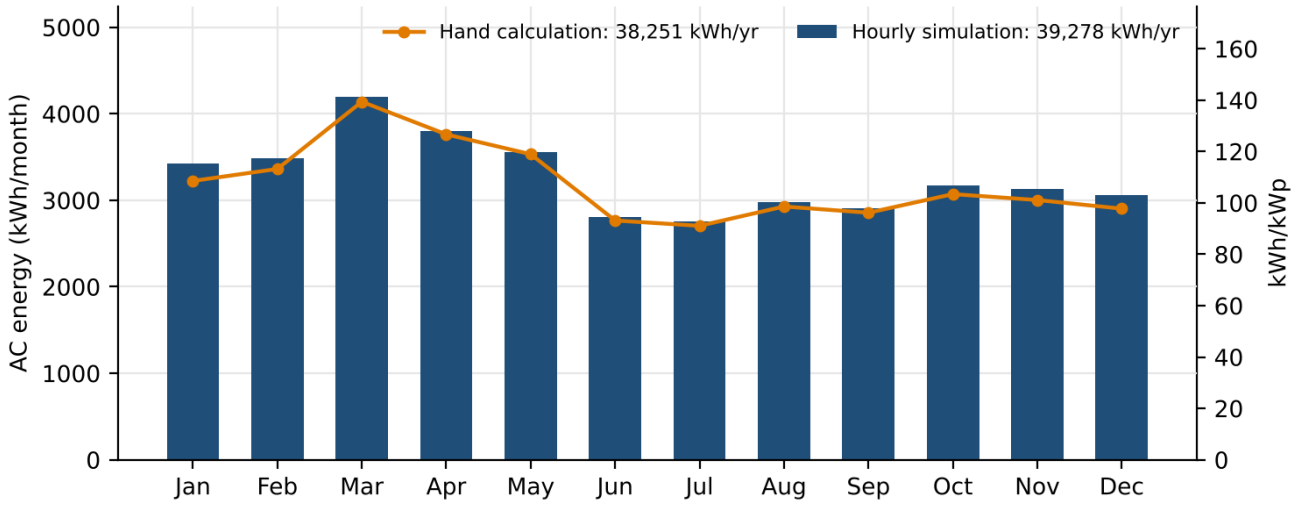
Check	Calculation	Result	Limit	Status
Max string voltage (8 C)	18 x 49.6 V x [1 + (-0.27%/C)(8 - 25)]	934 V	max 1100 V	OK
Min MPP voltage (cell 65 C)	18 x 41.9 V x [1 + (-0.35%/C)(65 - 25)]	649 V	min 200 V	OK
MPP voltage at STC	18 x 41.9 V	754 V	max 1000 V	OK
MPPT input current	2 string(s) x 13.13 A	26.3 A	max 32 A	OK
DC/AC ratio	29.70 kWp / 25.0 kW	1.19	1.0 - 1.3	OK



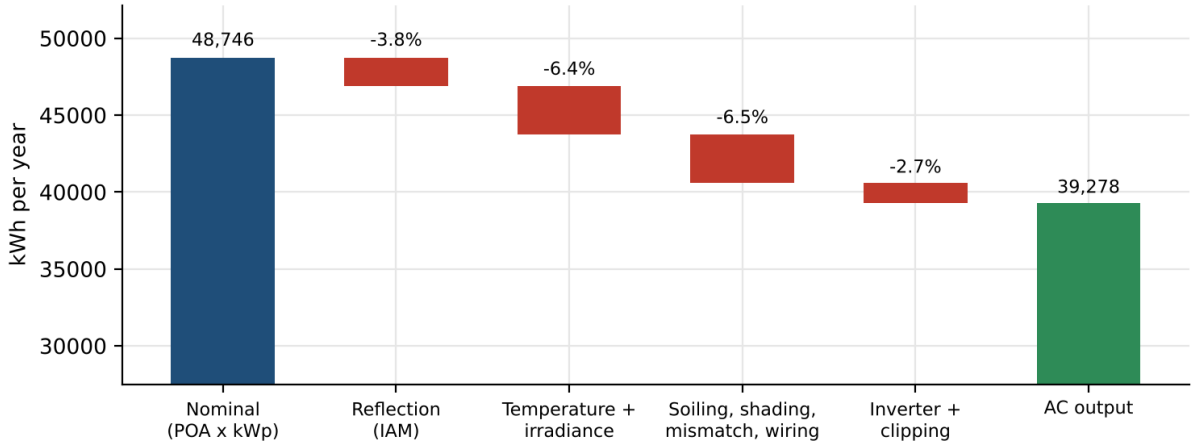
Single-line diagram, drawing PV-04.

PV and grid both feed the AC DB; chargers draw PV first, grid supplies any shortfall (44 kW chargers > 25 kW PV). Smart charging can shift EV sessions into solar hours. Steel canopy bonded to earth; ratings to be verified with datasheets.

4. Energy simulation

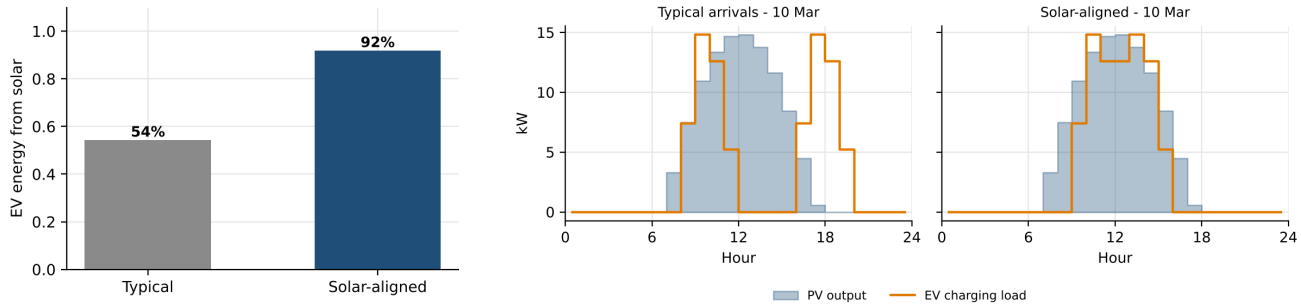


Monthly AC energy: hourly simulation 39,278 kWh/yr against a hand calculation of 38,251 kWh/yr (+2.7%). The hand calculation uses a fixed PR of 0.8; the simulation resolves each loss hour by hour.



Annual loss breakdown. Temperature and low-irradiance losses are 6.6% of effective irradiation; inverter clipping is 142 kWh/yr; peak AC output 25.0 kW.

5. EV charging analysis



Share of EV charging energy supplied by the canopy (left) and a working day in March for both schedules (right).

Schedule (20 kWh sessions at 7.4 kW)	EV energy	From solar	From grid	Solar share
Typical arrivals (08, 09, 16, 17 h)	22,960 kWh	12,466 kWh	10,494 kWh	54%
Solar-aligned (09, 10, 12, 13 h)	22,960 kWh	21,092 kWh	1,868 kWh	92%

With typical arrival times, the canopy supplies 54% of the charging energy; moving the same sessions into solar hours raises this to 92% and cuts grid energy for charging from 10,494 to 1,868 kWh/yr. Surplus PV is used by campus loads. This is the effect that a solar-aware charging scheduler aims to capture.

6. Bill of quantities

#	Item	Specification	Qty
1	PV module	550 Wp mono half-cell, IEC 61215/61730	54 nos
2	String inverter	25 kW, 3-phase, 2 MPPT, IEC 62109/61727	1 nos
3	Carport steel structure	HDG steel, single slope 10 deg, 12 bays, 2.7 m clear height	1 lot
4	Column foundations	RCC pad footing per structural design	10 nos
5	DC combiner box	3-string, isolator, Type II SPD	1 set
6	DC solar cable	4 mm2 PV1-F (3 strings x 2 x 50 m)	300 m
7	AC cables	4C x 16, 4C x 25, 4C x 35 mm2 Cu XLPE	1 lot
8	AC DB	MCCB 50 A 4P, RCD, SPD, sub-meter	1 set
9	EV charger	22 kW AC, Type 2, OCPP, RFID	2 nos
10	EV charging panel	2 x MCB 40 A 4P + RCD Type A/B	1 set
11	Earthing + lightning protection	Earth pits, bonding of canopy, air terminals	1 set
12	Under-canopy LED lights	IP65, 40 W	12 nos
13	Installation, testing & commissioning	Labour, testing per IEC 62446	1 lot

Project 5: 8.25 kWp Solar Irrigation Pump (7.5 kW) - Rajshahi

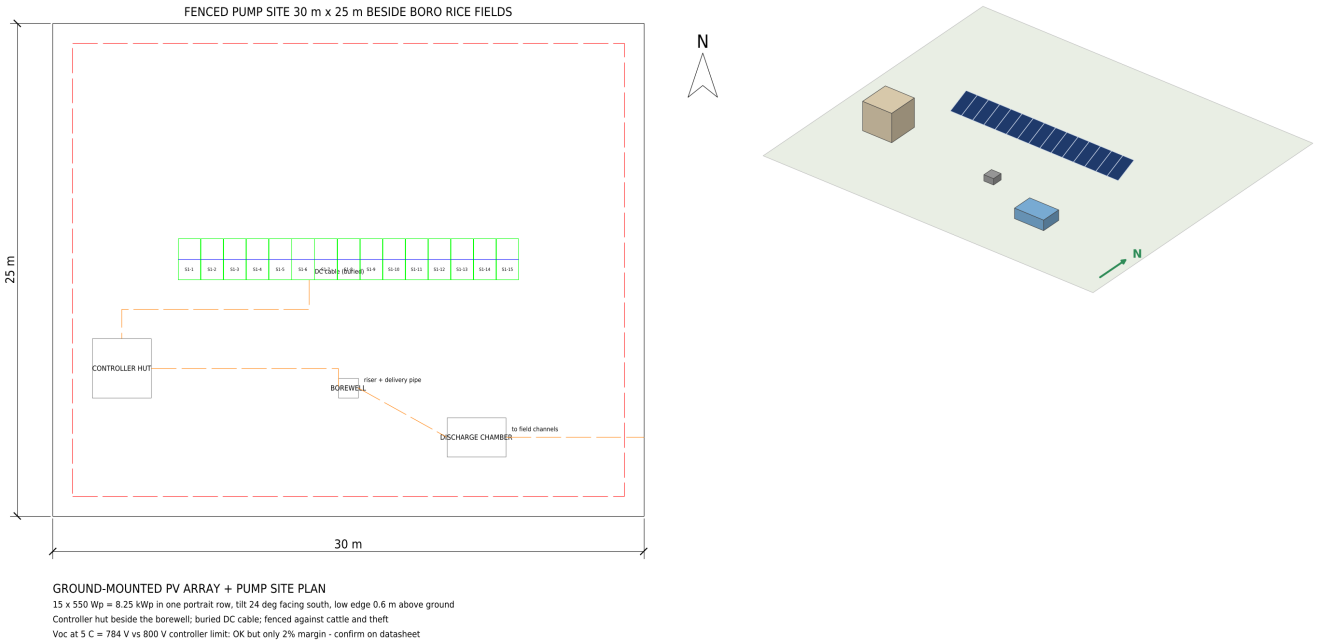
A stand-alone solar pumping system for Boro rice irrigation in Rajshahi. A 7.5 kW submersible pump is driven directly by a solar pump controller (VFD with MPPT); water storage in the discharge chamber replaces battery storage.

8.25 kWp	11,263	1,365	0.80	55.9k m3
15 x 550 Wp	kWh per year (simulated)	kWh/kWp per year	performance ratio	water per year

1. Site and design basis

Parameter	Value
Location	Rajshahi, Bangladesh (24.37 N, 88.6 E)
Site	Fenced pump site 30 m x 25 m beside boro rice fields
Orientation	Tilt 24 deg, azimuth 0 deg (south); tilt = latitude (24 deg); irrigation demand peaks in dry season (Jan-Apr)
PV array	15 x 550 Wp mono half-cell = 8.25 kWp; 1 string(s) of 15
Inverter / controller	7.5 kW solar pump controller (VFD with MPPT), 3-ph 380 V output
Design temperatures	Minimum ambient 5 C; maximum cell 70 C
Weather data	Meteonorm 9.0 monthly values (Dhaka region), hourly synthetic year

2. Array layout and 3D model

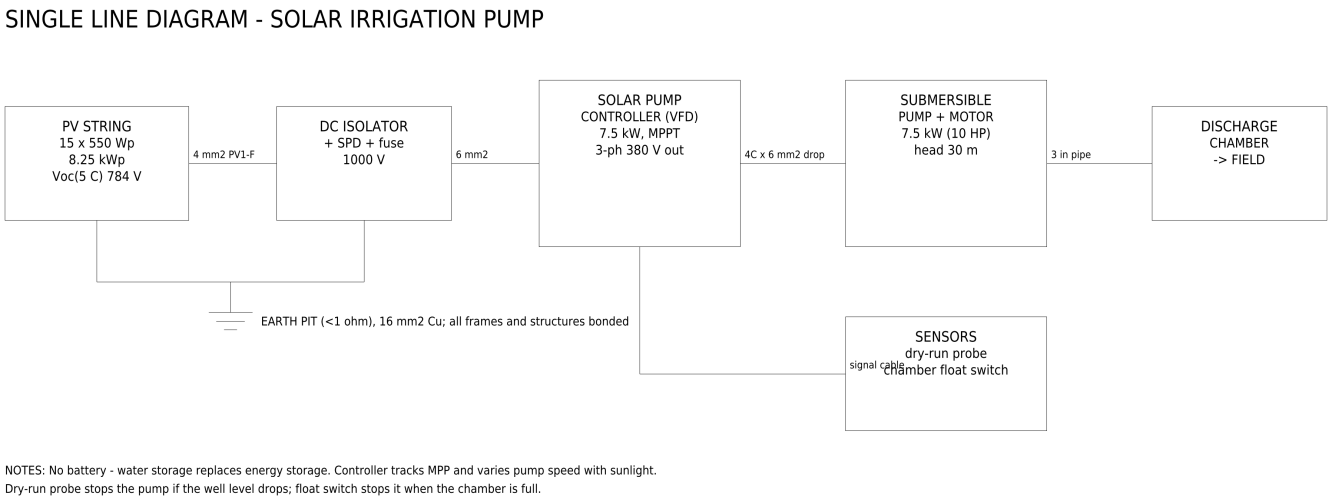


Layout plan from drawing PV-05 (left; string groups in separate colours) and 3D site model (right).

15 x 550 Wp = 8.25 kWp in one portrait row, tilt 24 deg facing south, low edge 0.6 m above ground. Controller hut beside the borewell; buried DC cable; fenced against cattle and theft. Voc at 5 C = 784 V vs 800 V controller limit: OK but only 2% margin - confirm on datasheet.

3. Electrical design

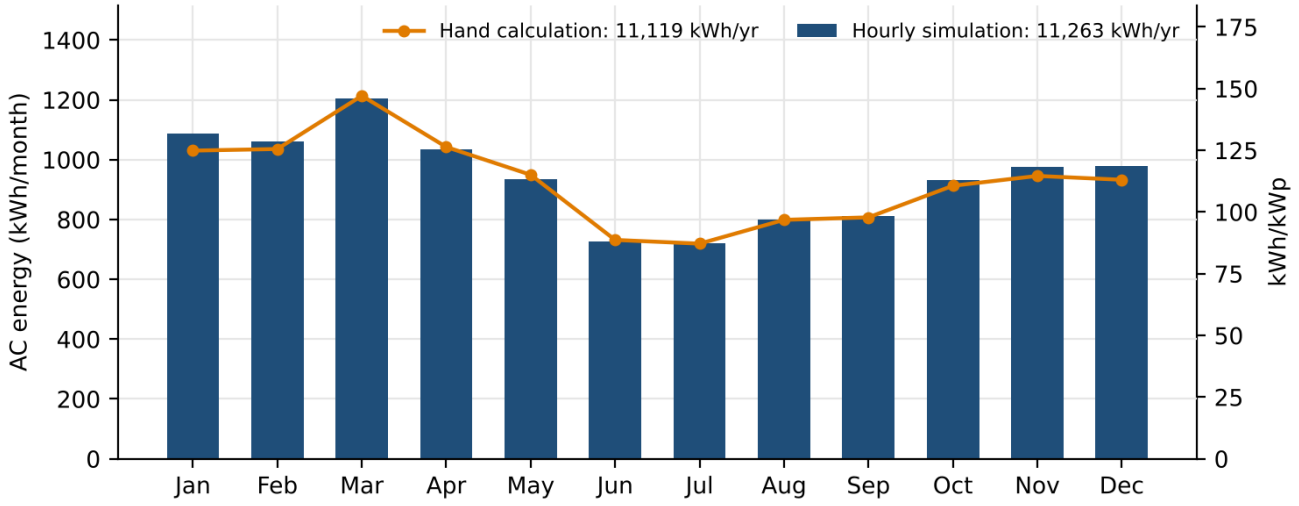
Check	Calculation	Result	Limit	Status
Max string voltage (5 C)	15 x 49.6 V x [1 + (-0.27%/C)(5 - 25)]	784 V	max 800 V	OK
Min MPP voltage (cell 70 C)	15 x 41.9 V x [1 + (-0.35%/C)(70 - 25)]	530 V	min 450 V	OK
MPP voltage at STC	15 x 41.9 V	628 V	max 750 V	OK
MPPT input current	1 string(s) x 13.13 A	13.1 A	max 20 A	OK
DC/AC ratio	8.25 kWp / 7.5 kW	1.10	1.0 - 1.5	OK



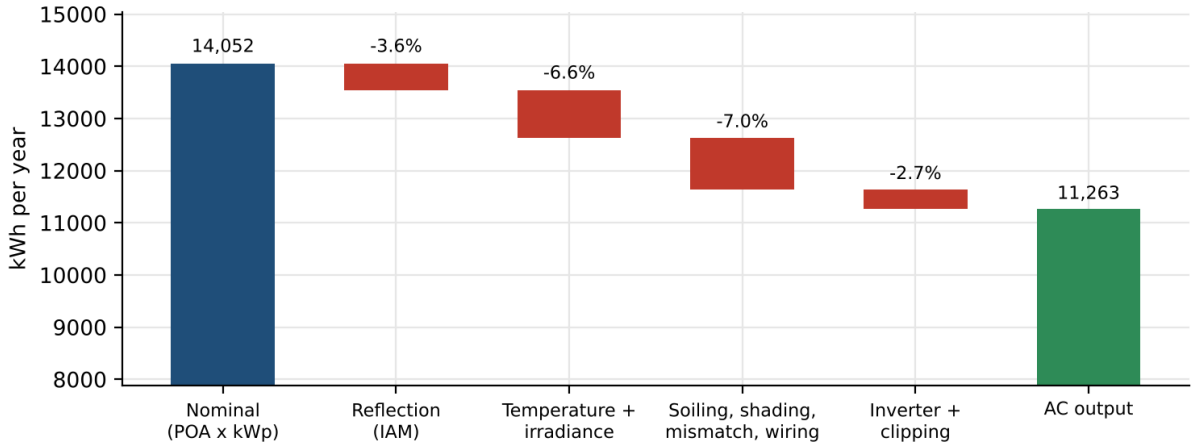
Single-line diagram, drawing PV-05.

No battery - water storage replaces energy storage. Controller tracks MPP and varies pump speed with sunlight. Dry-run probe stops the pump if the well level drops; float switch stops it when the chamber is full.

4. Energy simulation

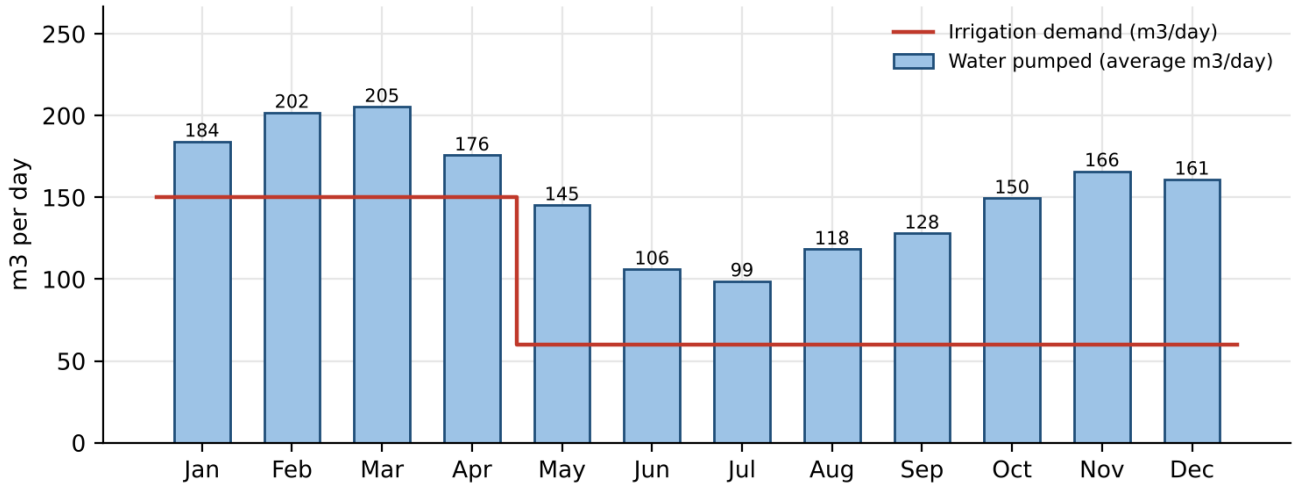


Monthly AC energy: hourly simulation 11,263 kWh/yr against a hand calculation of 11,119 kWh/yr (+1.3%). The hand calculation uses a fixed PR of 0.8; the simulation resolves each loss hour by hour.



Annual loss breakdown. Temperature and low-irradiance losses are 6.8% of effective irradiation; inverter clipping is 45 kWh/yr; peak AC output 7.0 kW.

5. Water delivery



Average daily water delivery at 30 m total dynamic head against the irrigation demand (150 m3/day in the Boro season, 60 m3/day otherwise).

The system pumps 55,853 m3 per year over 2,935 operating hours, with a peak flow of 41 m3/h. In the Boro season (January-April) it delivers 176-205 m3/day on average and meets the 150 m3/day demand on 70%-100% of days, enough for about 1.9 ha of rice at a peak water need of 8 mm/day. The string open-circuit voltage at 5 C (784 V) is within the controller's 800 V limit with a 2% margin, so the controller rating must be confirmed against the selected datasheet.

6. Bill of quantities

#	Item	Specification	Qty
1	PV module	550 Wp mono half-cell, IEC 61215/61730	15 nos
2	Solar pump controller	7.5 kW VFD with MPPT, 3-ph 380 V output, IP65	1 nos
3	Submersible pump set	7.5 kW (10 HP) stainless steel, 3-ph, head 30-40 m	1 set
4	Ground mounting structure	HDG steel, 24 deg, concrete piers	8.25 kWp
5	DC isolator + SPD + fuse	1000 V	1 set
6	DC solar cable	4 mm2 PV1-F, buried in conduit	80 m
7	Pump drop cable	4C x 6 mm2 flat submersible cable	50 m
8	Riser + delivery pipe	3 in uPVC / GI	40 m
9	Sensors	Dry-run probe + float switch	1 set
10	Controller hut + discharge chamber	Brick/RCC civil works	1 lot
11	Fencing	GI wire mesh fence with gate	110 m
12	Earthing + lightning arrester	Earth pits + air terminal	1 set
13	Installation, testing & commissioning	Labour, flow test	1 lot

Appendix A - PVsyst parameter sets

Parameter sets for reproducing each design in PVsyst 8.

Project 2: 49.5 kWp industrial rooftop, Gazipur

Step	Parameter	Value
1. Project	Project type	Grid-connected
1. Project	Site / weather	Gazipur: 23.99 N, 90.42 E, UTC+6; Meteonorm 9.0 monthly data, synthetic hourly year
2. Orientation	Field type / tilt / azimuth	Fixed tilted plane, tilt 5 deg, azimuth 0 deg (south)
3. System	Planned power	49.5 kWp
3. System	PV module	Mono 550 Wp half-cell
3. System	Inverter	20 kW three-phase, 2 MPPT, 2 units, Multi-MPPT
3. System	Array design	15 modules in series x 6 strings, Pnom ratio 1.24
4. Detailed losses	Thermal	Semi-integrated / close roof, Uc 20 W/m2K (flush on metal roof)
4. Detailed losses	Soiling	4% (industrial dust, Gazipur)
5. Near shadings	3D scene	Building 60x30x9 m; 3 PV tables (30 x 1 portrait, 5 deg) at z = 9.15 m; 4 ventilator boxes 1x1x1.2 m; Linear shadings
6. Target	Specific yield / PR	1,250-1,350 kWh/kWp/yr, PR 0.76-0.82

Project 3: 4.95 kWp hybrid + battery, Uttara

Step	Parameter	Value
1. Project	Project type	Grid-connected
1. Project	Site / weather	Uttara, Dhaka: 23.87 N, 90.4 E, UTC+6; Meteonorm 9.0 monthly data, synthetic hourly year
2. Orientation	Tilt / azimuth	Fixed tilted plane, 15 deg, azimuth 0 deg
3. System	Storage	Li-ion (LFP) 51.2 V, 200 Ah; strategy: self-consumption
3. System	PV + inverter	9 x 550 Wp in 1 string; 5 kW single-phase hybrid inverter
3. System	User's needs	Household profile, 15 kWh/day with evening peak
4. Detailed losses	Thermal / soiling	Free mounted (Uc 29 W/m2K); soiling 3%
5. Near shadings	3D scene	House 12x9x12 m; 2 PV tables (5 and 4 modules, 15 deg) at z = 12.3 m; tank 1.8x1.5x1.5; stair room 2.5x2x2.5
6. Target	Production / indicators	6.0-6.7 MWh/yr; self-consumption and solar fraction

Project 4: 29.7 kWp solar carport + EV charging

Step	Parameter	Value
1. Project	Project type	Grid-connected
1. Project	Site / weather	AUST, Tejgaon, Dhaka: 23.76 N, 90.41 E, UTC+6; Meteonorm 9.0 monthly data, synthetic hourly year
2. Orientation	Tilt / azimuth	Fixed tilted plane, 10 deg, azimuth 0 deg
3. System	PV + inverter	18 modules in series x 3 strings; 25 kW three-phase inverter, Multi-MPPT
3. System	User's needs	EV charging profile: 4 sessions x 20 kWh on working days
4. Detailed losses	Thermal / soiling	Free mounted with air circulation (Uc 29 W/m2K); soiling 3%
5. Near shadings	3D scene	Two PV tables (27 x 1 portrait) at z = 2.7 m and 3.1 m, tilt 10 deg
6. Target	Specific yield / PR	1,250-1,350 kWh/kWp/yr (37-40 MWh/yr), PR ~0.80

Project 5: 8.25 kWp solar irrigation pump, Rajshahi

Step	Parameter	Value
1. Project	Project type	Pumping system: deep well to storage
1. Project	Site	Rajshahi: 24.37 N, 88.60 E, UTC+6; Meteonorm 9.0 monthly data
2. Orientation	Tilt / azimuth	Fixed tilted plane, 24 deg, azimuth 0 deg
3. Hydraulic circuit	Well	Static level 18 m, specific drawdown ~0.5 m/(m3/h), pump depth 30 m
3. Hydraulic circuit	Storage	Discharge chamber modelled as a 20 m3 tank, outlet at ground level
3. Water needs	Daily demand	150 m3/day in Jan-Apr (Boro), 60 m3/day other months
4. System	Pump	Submersible centrifugal pump, 7.5 kW, 3-phase
4. System	Controller	MPPT controller with AC (inverter) output for 3-phase motor
4. System	PV array	15 x 550 Wp in series, 1 string
5. Target	Water delivered	140-200 m3/day in dry months; missing-water fraction

Appendix B - Software and code

All drawings, workbooks, 3D models, simulations and figures in this report are generated from one Python code base (github.com/radit057/Solar-PV), so a change to a design input propagates to every output.

Module	Purpose
projects.py	Design definitions for each project: site, array, electrical, BOQ, 3D geometry, simulation specification
solarpv/design.py	Module data, plan geometry of tilted modules, winter-solstice row-pitch calculation
solarpv/cad.py, drawing.py	AutoCAD DXF generation: layers per string, dimensions, single-line diagram, title block
solarpv/excel.py, sheets.py	Excel design workbook with live formulas: string sizing, energy estimate, BOQ, battery/EV/water sheets
solarpv/sketchup.py	SketchUp Ruby script generator: site model, PV components, Dhaka geo-location, shadow scenes
solarpv/weather.py	Hourly weather: PVGIS TMY or synthetic year from Meteonorm monthly data
solarpv/simulation.py	pvlib hourly model chain and loss breakdown
solarpv/systems.py	Battery dispatch, EV charging coverage, solar pump flow model
solarpv/figures.py, render.py	Report figures and isometric 3D renders
build_all.py, report/build_report.py	Build pipeline and this report